

Alex Chen

Software Engineering Student | Full-Stack Developer | AI Application Builder

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PROFILE

Software engineering student with strong experience in Java web development, database design, AI-assisted applications, and team-based software projects. Built multiple full-stack systems using JSP/Servlet, Spring Boot, React, MySQL, REST APIs, and cloud deployment workflows. Experienced in competitive programming, hackathons, and software engineering competitions.

EDUCATION

Beijing University of Posts and Telecommunications - BSc Software Engineering, 2023-2027

Relevant coursework: Software Engineering, Data Structures, Object-Oriented Programming, Database Systems, Web Development, Operating Systems, Computer Networks, Machine Learning Basics.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript, SQL, C/C++
Backend	JSP/Servlet, Spring Boot, REST APIs, Maven, JUnit, Mockito, Tomcat
Frontend	HTML, CSS, React, Vue basics, responsive UI, form validation
Data & AI	MySQL, PostgreSQL, file-based persistence, JSON parsing, prompt engineering, LLM API integration
Tools	Git, GitHub, Linux shell, Docker basics, Postman, Figma, IntelliJ IDEA, VS Code

SOFTWARE ENGINEERING PROJECTS

TA Recruitment Management System - Java Servlet/JSP, Tomcat, JUnit

- ? Designed a role-based recruitment platform for TA, Module Organizer, and Admin users with separate entry points, authentication flows, and permission checks.
- ? Implemented file-based persistence for users, courses, deadlines, resumes, and course-specific application states.
- ? Built application deadline handling, resume upload, review status tracking, and unit tests covering controller, service, model, and store layers.
- ? Refactored controller-heavy code into service interfaces and implementation classes to reduce coupling and improve testability.

AI-Assisted Job Application Form Generator - Java, REST API, JSON

- ? Built a prototype that transforms a global resume and course job description into a structured job-specific application form.
- ? Designed prompt templates for extracting name, education, skills, relevant experience, project experience, course fit, and improvement feedback.
- ? Implemented editable draft and final submission workflows so applicants can revise AI-generated forms before deadline.

Campus Event Recommendation Platform - Spring Boot, React, MySQL

- ? Developed REST APIs for event publishing, user registration, favorites, search, and personalized recommendation ranking.
- ? Designed normalized database tables and optimized keyword/category filtering queries for improved response time.
- ? Implemented frontend components for event cards, user dashboard, admin moderation, and responsive mobile layouts.

Online Judge Mini System - Python, Docker, PostgreSQL

- ? Created a sandboxed code execution workflow for programming assignments using isolated Docker containers.
- ? Implemented submission queue, verdict storage, test case management, and teacher-facing result analytics.
- ? Added plagiarism warning signals based on output similarity and code metadata comparison.

COMPETITIONS & AWARDS

- ? Second Prize, University Software Engineering Innovation Challenge - led a 5-member team building an AI-powered course assistant prototype.
- ? Finalist, BUPT Hackathon 2025 - developed a real-time study group matching web app using Spring Boot and React in 36 hours.
- ? Third Prize, ACM-ICPC Provincial Invitational Practice Contest - solved graph, dynamic programming, and greedy algorithm problems under time pressure.

- ? Outstanding Project Award, Database Systems Course Design - built a library management system with transaction-safe borrowing and admin analytics.
- ? Best Team Collaboration Award, Software Engineering Group Project - responsible for backend architecture, testing strategy, and release integration.

ENGINEERING EXPERIENCE

Backend Developer Intern, Cloud Learning Lab - Jun 2025 to Aug 2025

- ? Implemented Java REST endpoints for assignment submission, notification delivery, and user activity logs.
- ? Wrote JUnit tests and integration test cases that improved backend regression coverage for core learning workflows.
- ? Collaborated with frontend developers to define API contracts and debug cross-origin/session issues.

TEACHING & COLLABORATION

- ? Helped classmates debug Java Servlet, Git merge conflict, and database schema issues during group coursework.
- ? Prepared short tutorials on MVC architecture, unit testing with Mockito, and REST API design for study group sessions.
- ? Comfortable reading legacy code, documenting workflows, reviewing pull requests, and explaining technical trade-offs clearly.

SELECTED STRENGTHS FOR TA ROLES

- ? Strong foundation in Java, web backend development, testing, debugging, and software project organization.
- ? Able to support lab sessions involving JSP/Servlet, object-oriented design, Git workflows, database design, and API integration.
- ? Experienced in turning ambiguous requirements into structured tasks, tests, and deliverable software increments.